HOTEL ROOM BOOKING SYSTEM

(PBCST304 -OBJECT ORIENTED PROGRAMMING)

Project Report

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OCTOBER 2025

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Marian Engineering College

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CERTIFICATE

This is to certify that this project work report entitled '**HOTEL ROOM BOOKING SYSTEM**' submitted by **PARVATHY J KUMAR (MCE24AIM046)**, **SANDRA J S (MCE24AIM056), SREESHNA VM (MCE24AIM060)**, and **YANIK SAN VARGHESE (MCE24AIM061)** of Third Semester, B.Tech in Artificial Intelligence and Machine Learning has been successfully carried out under our supervision and guidance in partial fulfillment of the requirements of the project for the subject PBCST304 – Object Oriented Programming, prescribed by APJ Abdul Kalam Technological University, Kerala. It is further certified that this work is a bona fide record of the project carried out by the above students during the academic year **2025–2026**.

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DECLARATION

We hereby declare that the microproject report HOTEL ROOM BOOKING submitted for the partial fulfilment of the requirements of the Third Semester Project for the subject PBCST304 – Object Oriented Programming, prescribed by APJ Abdul Kalam Technological University, Kerala is a bona fide work done by us under the supervision of Ms. Renetha J B, Assistant Professor, CSE.

The submission represents our ideas in our own words and where ideas or words of others have not been included. We have adequately and accurately cited and referred the original sources.

We also declare that we have adhered to the ethics and integrity and have not misinterpreted or fabricated any data or idea or fact or source in our submission .We understand that any violation of the above will be a case for disciplinary action by the institution and/or the University and can also evoke penal actions from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

The **Hotel Room Booking System** is a Java-based application developed to simplify and automate the process of hotel room reservations. The system enables users to search for hotels, view available rooms, check amenities, and make bookings through an easy-to-use graphical interface. It also allows administrators to manage hotel information, room details, pricing, and booking records efficiently.

By integrating **Java Swing** for the user interface and **MySQL** for the database, the system ensures secure data storage, quick access, and smooth interaction between users and the database. This computerized approach eliminates the limitations of manual booking systems such as data inconsistency, errors, and time delays.

This system provides a centralized platform that enhances operational efficiency, improves customer satisfaction, and supports real-time management of hotel operations. It serves as a reliable and scalable solution for modernizing hotel management through technology.

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LIST OF ABBREVIATIONS

Abbreviation	**Expansion**
SQL	Structured Query Language.
IDE	Integrated Development Environment
AWT	Abstract Window Toolkit
HRBS	Hotel Room Booking System
***	***

Chapter 1

INTRODUCTION

In the modern era, technology has transformed the hospitality industry by redefining how travelers search, book, and experience accommodation. With the increasing dependency on digital platforms, hotel booking and management systems have become an essential component of travel and tourism infrastructure. The demand for convenience, transparency, and efficiency has prompted both customers and hoteliers to move away from traditional manual booking processes toward online and automated solutions.

Traditionally, hotel room booking was managed through manual registers or over-the-counter reservations, which were prone to errors, data loss, overbooking, and communication gaps. This manual system often resulted in customer dissatisfaction and inefficient room utilization. In addition, the absence of a centralized platform made it difficult for hotel management to maintain accurate records of room availability, guest details, pricing, and amenities.

Hotel Room Booking System

This system is designed to address these limitations by providing an automated and user-friendly Java-based platform that allows users to browse hotels, check room availability, view amenities, and make bookings conveniently. For administrators and hotel owners, the system simplifies management by enabling real-time updates of room details, pricing, and booking status through a secure backend.

By incorporating database connectivity using MySQL and an intuitive graphical interface through Java Swing, StayZy ensures efficient data handling and smooth user experience. The system not only enhances operational efficiency but also improves customer satisfaction through accuracy, accessibility, and transparency.

In a world where time and convenience are paramount, **StayZy** provides a technological bridge between travelers and hotels, streamlining the booking process while maintaining reliability and scalability. It represents a significant step toward digital transformation in hotel management and reservation systems.

Chapter 2

AIM AND OBJECTIVES

The aim of the StayZy Hotel Booking Management System is to develop a computerized, efficient, and user-friendly platform that facilitates seamless room booking, management, and record maintenance for hotels and users alike. The system aims to centralize hotel information, reduce manual work, and ensure real-time communication between customers and hotel administrators.

The system ensures that all hotel-related data — including hotel details, room types, pricing, amenities, and booking status — are stored in a **centralized MySQL database**, allowing both guests and hotel staff to access accurate and updated information instantly. This approach minimizes human error, prevents double booking, and increases operational transparency.

The Objectives of the proposed system are as follows:

i. To design a user-friendly Java-based interface for hotel room booking and management.

ii. To maintain centralized and secure hotel and user data through MySQL database integration.

iii. To enable users to view hotel details and amenities in real-time.

iv. To automate the process of booking, cancellation, and payment confirmation.

v. To reduce paperwork and manual errors associated with hotel management.

vii. To ensure scalability for integrating multiple hotels within one platform.

Chapter 3

REQUIREMENT SPECIFICATION

3.1 SOFTWARE REQUIREMENTS

- **Operating System:** Windows 10 or higher / Linux / macOS
- **Programming Language:** Java
- **Development Environment:** Eclipse IDE
- **Database:** MySQL Server 5.7 or higher
- **Connectivity:** JDBC (Java Database Connectivity) driver
- **User Interface:** Java Swing / AWT
- **Web Server / Workbench:** XAMPP (includes Apache Server and MySQL)
- **Other Tools:** MySQL Workbench for database management

3.2 **HARDWARE REQUIREMENTS**

- Processor: Minimum Pentium III – 900 MHz or higher
- RAM: 512 MB (minimum), 1 GB or higher recommended
- Hard Disk: Minimum 500 MB free space, 4 GB recommended
- Display: Standard VGA Monitor or higher
- Input Devices: Keyboard and Mouse.

Chapter 4

DESIGN DOCUMENT

4.2 DATABASE DESIGN

A database is the core component of any management system, serving as the central repository for storing and retrieving information efficiently. In the **Hotel Booking System**, the database stores details related to hotels, rooms, users, and bookings. Since the accuracy and reliability of the database directly affect system performance, it is essential to test the database to ensure it meets the **ACID properties** — **Atomicity, Consistency, Isolation, and Durability**.

Database testing helps to identify and eliminate integration, implementation, and data handling errors that could negatively impact the system's operation. Testing ensures that data is correctly stored, retrieved, and updated during various operations such as user registration, room booking, and payment confirmation.

Below are some screenshots and descriptions of the key tables used in the proposed system.

ADMIN REGISTRATION TABLE

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
userid	int	NO	PRI	NULL	auto_increment
firstname	varchar(50)	NO		NULL	
lastname	varchar(50)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
phone	varchar(15)	NO		NULL	
alternate_phone	varchar(15)	YES		NULL	
dob	date	YES		NULL	
address	varchar(255)	YES		NULL	
city	varchar(100)	YES		NULL	
state	varchar(100)	YES		NULL	
country	varchar(100)	YES		NULL	
password	varchar(255)	NO		NULL	

Table 1 shows the **UserDetails (or Admin)** table, which stores registration information for users or administrators of the StayZy Hotel Booking System. The table maintains personal and login details that are essential for authentication and management purposes.

Each record is uniquely identified by the **UserID**, which is set as the **Primary Key** and is auto-incremented for each new user entry. The **email** field is marked as **unique**, ensuring no duplicate accounts are created

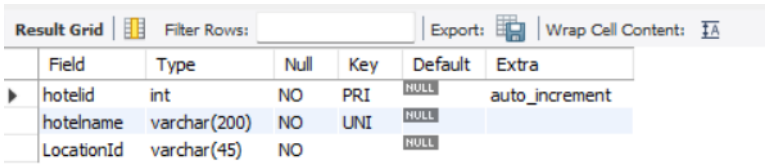
LOCATION DETAILS TABLE

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

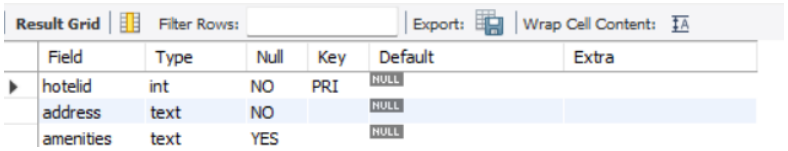
Field	Type	Null	Key	Default	Extra
LocationId	int	NO	PRI	NULL	auto_increment
LocationName	varchar(100)	YES	UNI	NULL	
State	varchar(45)	YES		NULL	

Table 2 shows the locationdetails table, stores information about hotel locations, including locationid, locationname, and state.

HOTEL DETAILS TABLE



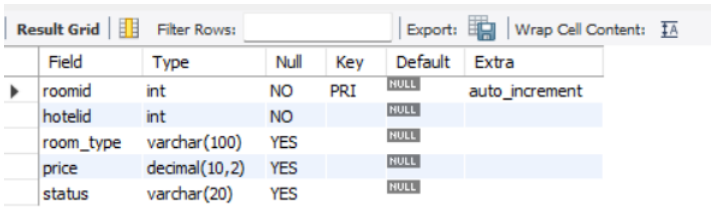
Field	Type	Null	Key	Default	Extra
hotelid	int	NO	PRI	NULL	auto_increment
hotelname	varchar(200)	NO	UNI	NULL	
LocationId	varchar(45)	NO		NULL	



Field	Type	Null	Key	Default	Extra
hotelid	int	NO	PRI	NULL	
address	text	NO		NULL	
amenities	text	YES		NULL	

Table 3 represents the **HotelDetails** table, which stores essential information about each hotel in the StayZy system. It contains the **hotelid** as the **Primary Key**, ensuring that each hotel entry is uniquely identified. The table also records the hotel's **address** and **amenities** to help users understand the facilities offered by each hotel.

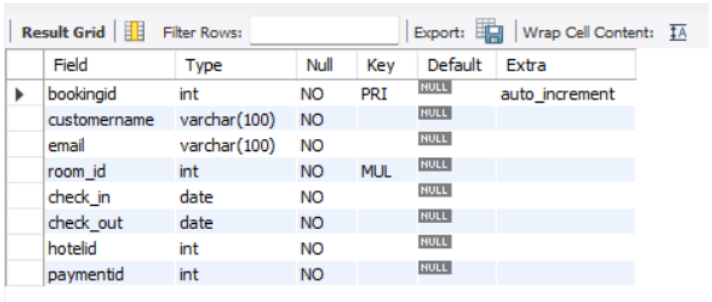
ROOM DETAILS TABLE



Field	Type	Null	Key	Default	Extra
roomid	int	NO	PRI	NULL	auto_increment
hotelid	int	NO		NULL	
room_type	varchar(100)	YES		NULL	
price	decimal(10,2)	YES		NULL	
status	varchar(20)	YES		NULL	

Table 4 represents the **RoomDetails** table, which stores essential information about the different types of rooms available in each hotel. It contains the **roomid** as the **Primary Key**, ensuring that each room type is uniquely identified. The table also includes the **hotelid** as a **foreign key** to link each room to its corresponding hotel.

BOOKING DETAILS TABLE



Field	Type	Null	Key	Default	Extra
bookingid	int	NO	PRI	NULL	auto_increment
customername	varchar(100)	NO		NULL	
email	varchar(100)	NO		NULL	
room_id	int	NO	MUL	NULL	
check_in	date	NO		NULL	
check_out	date	NO		NULL	
hotelid	int	NO		NULL	
paymentid	int	NO		NULL	

Table 5 represents the **BookingDetails** table, which records all booking transactions. It includes **bookingid** (primary key), **customername**, **emailid**, **room_id**, **check_in** , **check_out** , **hotelid**(connectivity)and **paymentid**

Chapter 5

SYSTEM IMPLEMENTATION

Programming language used includes the followings:

- Java

  Stayzy is developed using Java, leveraging object-oriented programming (OOP) principles such as classes, objects, inheritance, and encapsulation. This allows for modular, reusable, and maintainable code, making the system robust and scalable.

- Structured query language.

  The application uses SQL to manage hotel, user, and booking data efficiently. Structured queries allow for easy data storage, retrieval, and manipulation, ensuring seamless integration between the front-end and database.

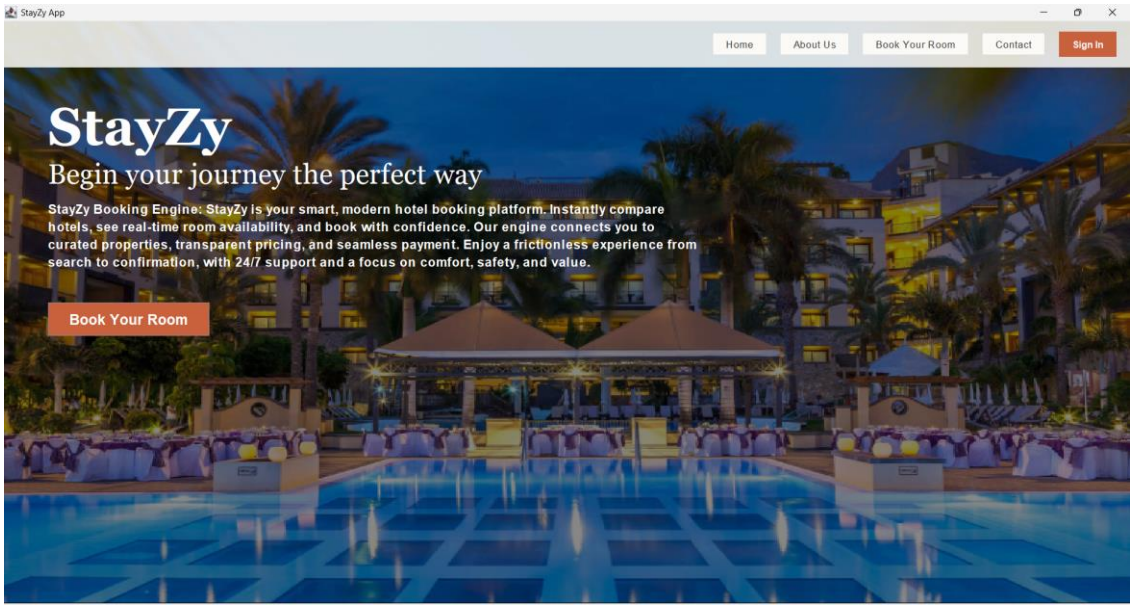
Chapter 6

SYSTEM OUTPUT/SCREENSHOT

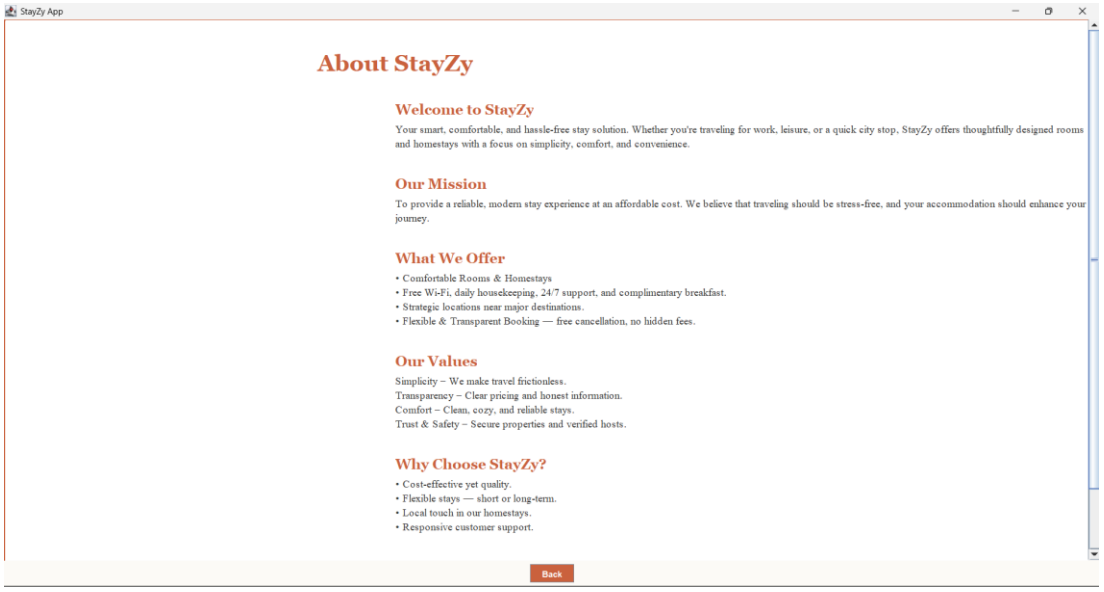
This involves testing the various modules of the system individually, integrating the various components into one, and testing the HRBS as a whole. The HRBS consists of various modules. Some of them are:

HOME PAGE:

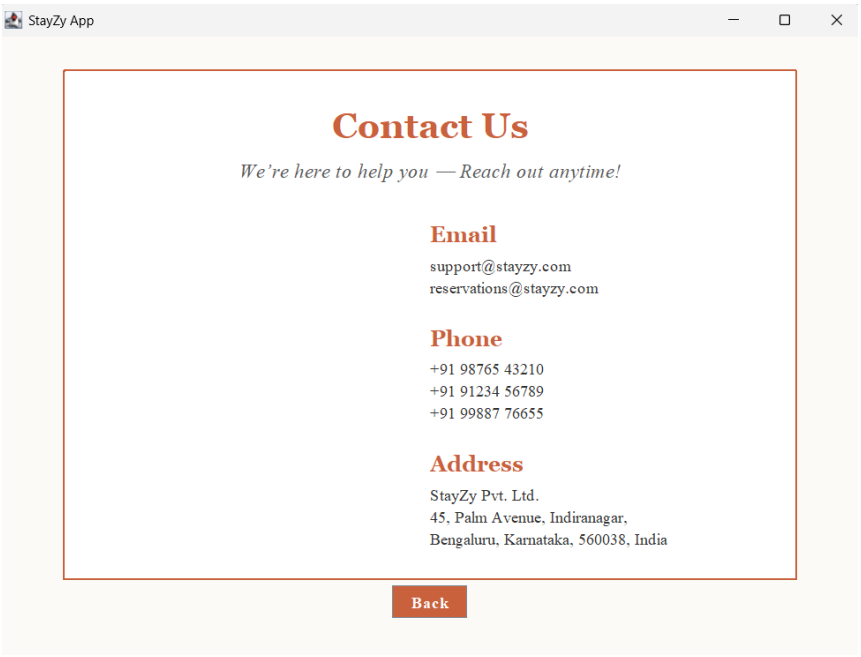
This is the home page of the Stayzy hotel room booking system. It contains navigation buttons such as Home, About Us, Book Your Room, and Contact.



ABOUT US PAGE:

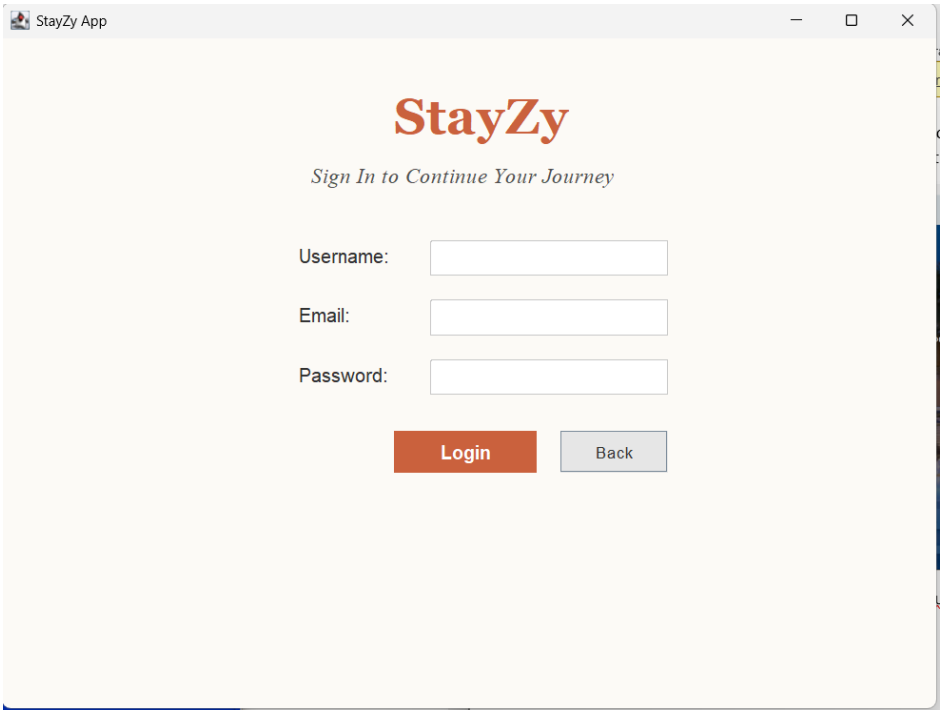


CONTACT PAGE:



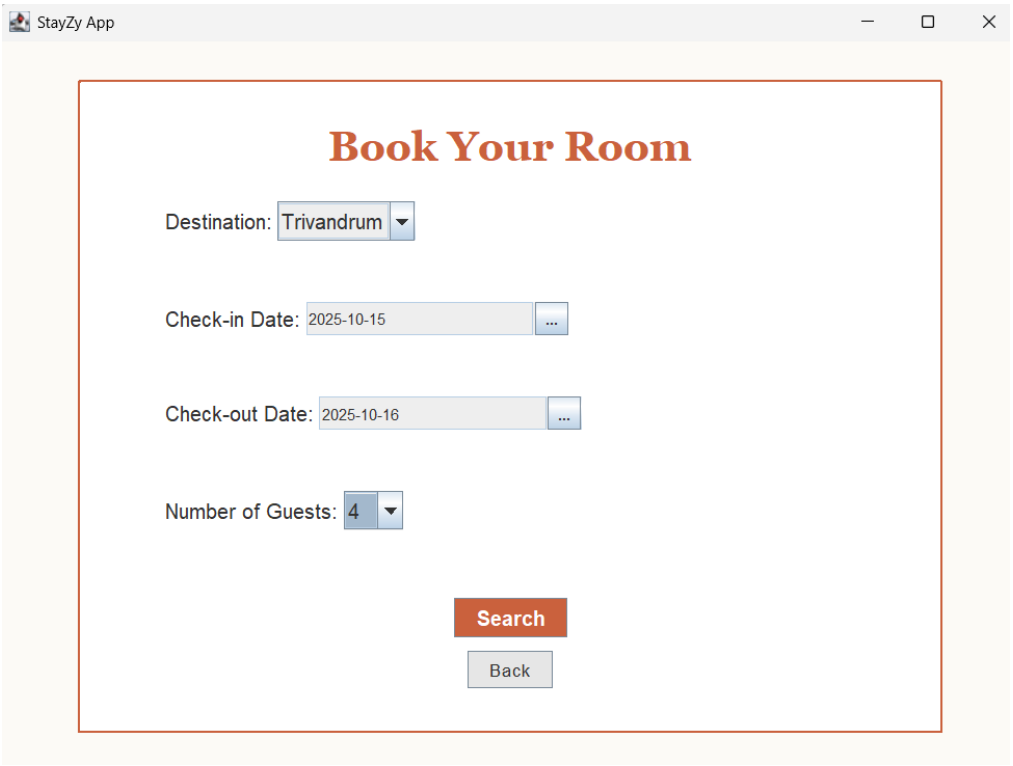
SIGN IN PAGE:

First-time users need to click the Sign In button. Enter your username, email, and password, then click Login to access your account. Further details will be asked after selecting hotel for booking.



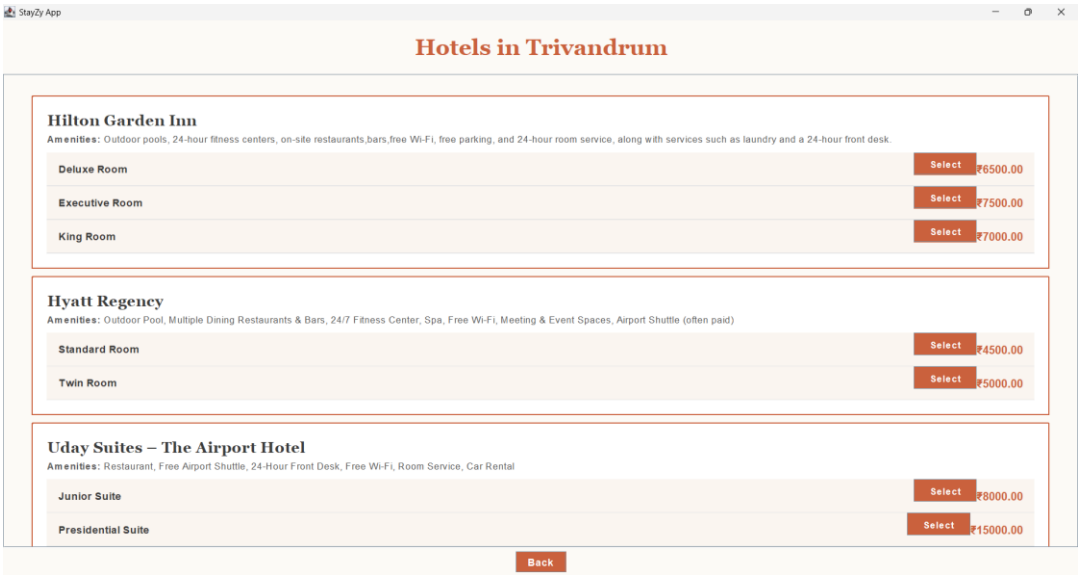
BOOK YOUR ROOM PAGE:

Select your desired location in Kerala, choose the check-in and check-out dates, specify the number of guests, and click **Search** to find available rooms

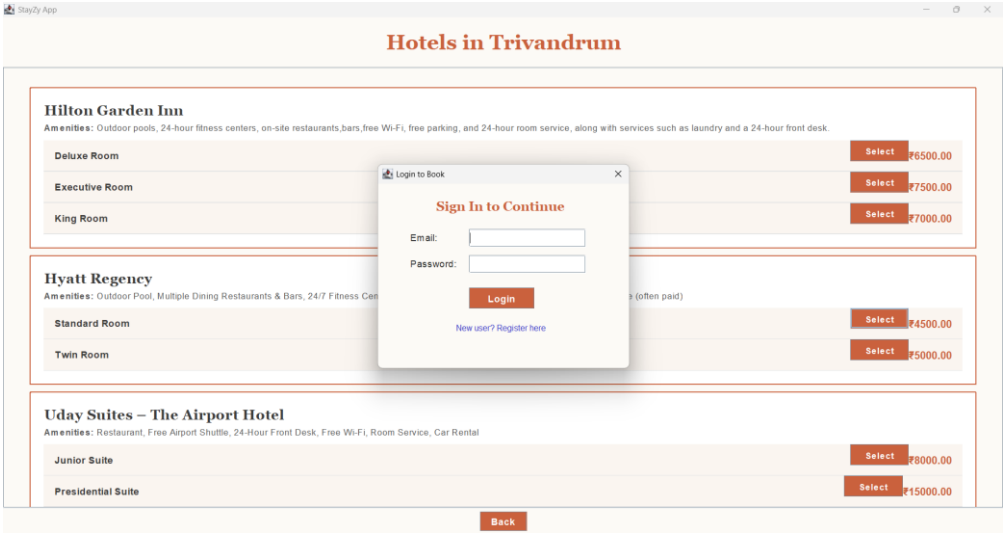


HOTEL LIST PAGE:

This page displays hotels available in Stayzy for the selected location, including room types and rates. Select a room and click **Select**.

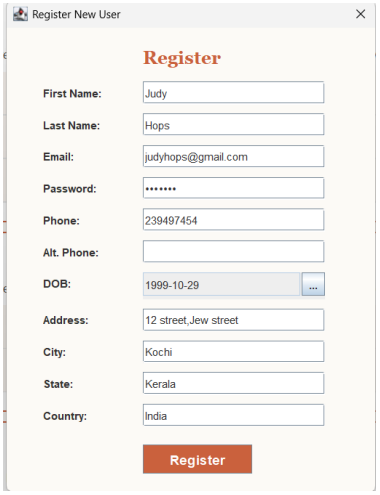


- If not signed in, click **New User**, fill in the registration details, and sign in.
- Returning users can log in using their email and password.



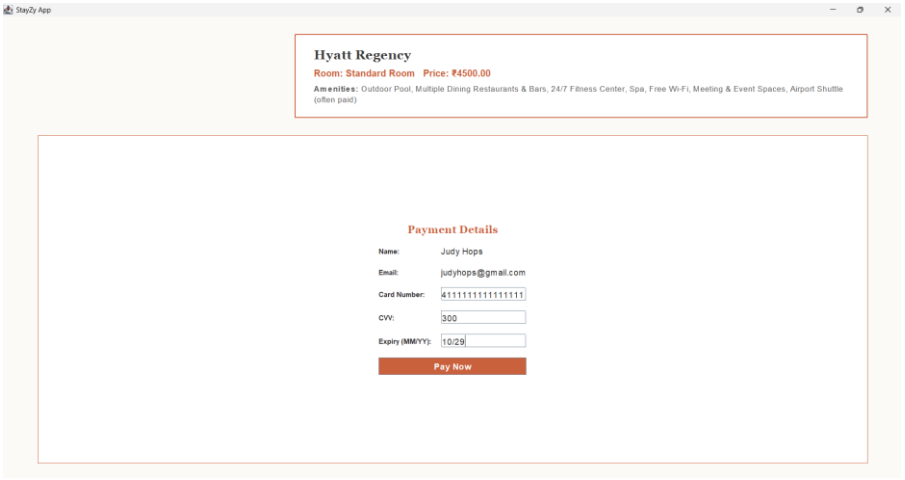
REGISTRATION PAGE:

New users can register by providing the required details, creating an account to proceed with booking.



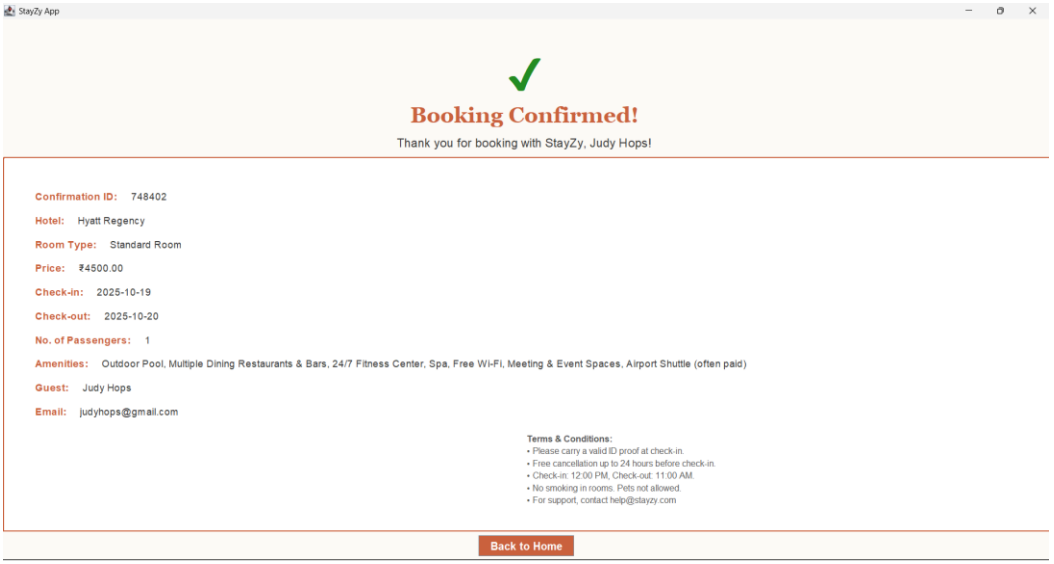
PAYMENT PAGE:

Enter card details to make the payment and click **Pay Now** to complete the transaction.



CONFIRMATION PAGE:

This page displays the booking confirmation along with all relevant details and terms & conditions.



Chapter 7

CODE LISTING

```
package Booking;
import javax.swing.*;
import javax.swing.border.*;
import java.awt.*;
import java.awt.event.*;
import org.jdatepicker.impl.*;
import java.text.ParseException;
import java.text.SimpleDateFormat;
import java.util.Calendar;
import java.util.Date;
import java.util.Properties;
import javax.swing.JFormattedTextField.AbstractFormatter;
import java.sql.*;
import java.util.ArrayList;

public class StayZyApp extends JFrame{

   JFrame frame;

   // --- DB Credentials (use these everywhere) ---
   private static final String DB_URL = "jdbc:mysql://localhost:3306/hotel";
   private static final String DB_USER = "root";
   private static final String DB_PASS = "root";
   Image backgroundImage =null;
   public StayZyApp() {
      frame = new JFrame("StayZy - Begin your journey the perfect way");
      frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
      frame.setSize(900, 600);
      frame.setLocationRelativeTo(null);
      showHomePage();
      frame.setVisible(true);
   }

   //--- Home Page ---
   private void showHomePage() {
   frame.getContentPane().removeAll();
   frame.setTitle("StayZy App");
```

```
frame.setSize(800, 600);
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
frame.setLocationRelativeTo(null);

// ---- Load Background Image Safely ----
java.net.URL imgUrl = getClass().getResource("/images/stayzy.jpg");
//Image backgroundImage = null;
if (imgUrl != null) {
   ImageIcon icon = new ImageIcon(imgUrl);
   backgroundImage = icon.getImage().getScaledInstance(800, 600, Image.SCALE_SMOOTH);
} else {
   System.err.println("✕ Background image not found at /images/stayzy.jpg");
}

// ---- Main Panel with Custom Paint ----
JPanel mainPanel = new JPanel(new BorderLayout()) {
   @Override
   protected void paintComponent(Graphics g) {
      super.paintComponent(g);
      if (backgroundImage != null) {
         g.drawImage(backgroundImage, 0, 0, getWidth(), getHeight(), this);
      }

      // Semi-transparent overlay for better text visibility
      Graphics2D g2d = (Graphics2D) g.create();
      g2d.setColor(new Color(0, 0, 0, 120)); // translucent black
      g2d.fillRect(0, 0, getWidth(), getHeight());
      g2d.dispose();
   }
};
mainPanel.setOpaque(false);

// ---- Top Navigation Bar ----
JPanel navBar = new JPanel(new FlowLayout(FlowLayout.RIGHT, 20, 15));
navBar.setBackground(new Color(252, 250, 246, 220));

JButton homeBtn = createNavButton("Home");
JButton aboutBtn = createNavButton("About Us");
JButton bookBtn = createNavButton("Book Your Room");
JButton contactBtn = createNavButton("Contact");
JButton signInBtn = new JButton("Sign In");
```

```
signInBtn.setBackground(new Color(202, 97, 61));
signInBtn.setForeground(Color.WHITE);
signInBtn.setFocusPainted(false);
signInBtn.setBorder(new EmptyBorder(8, 18, 8, 18));
signInBtn.setFont(new Font("SansSerif", Font.BOLD, 13));
signInBtn.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));

navBar.add(homeBtn);
navBar.add(aboutBtn);
navBar.add(bookBtn);
navBar.add(contactBtn);
navBar.add(signInBtn);
mainPanel.add(navBar, BorderLayout.NORTH);

// ---- Center Overlay ----
JPanel overlayPanel = new JPanel(null);
overlayPanel.setOpaque(false);
int contentLeft = 60;

// Title
JLabel title = new JLabel("StayZy");
title.setFont(new Font("Georgia", Font.BOLD, 72));
title.setForeground(Color.WHITE);
title.setBounds(contentLeft, 40, 500, 80);
overlayPanel.add(title);

// Subtitle
JLabel subtitle = new JLabel("Begin your journey the perfect way");
subtitle.setFont(new Font("Georgia", Font.PLAIN, 38));
subtitle.setForeground(Color.WHITE);
subtitle.setBounds(contentLeft, 120, 900, 50);
overlayPanel.add(subtitle);

// Description
JTextArea desc = new JTextArea(
    "StayZy Booking Engine: StayZy is your smart, modern hotel booking platform. " +
    "Instantly compare hotels, see real-time room availability, and book with confidence. " +
    "Our engine connects you to curated properties, transparent pricing, and seamless payment. " +
    "Enjoy a frictionless experience from search to confirmation, with 24/7 support and a focus on comfort, safety, and
value."
```

```
        );
        desc.setFont(new Font("SansSerif", Font.BOLD, 18));
        desc.setForeground(Color.WHITE);
        desc.setOpaque(false);
        desc.setEditable(false);
        desc.setLineWrap(true);
        desc.setWrapStyleWord(true);
        desc.setBounds(contentLeft, 180, 900, 120);
        desc.setBorder(null);
        overlayPanel.add(desc);

        // "Book Now" button
        JButton bookNowBtn = new JButton("Book Your Room");
        bookNowBtn.setFont(new Font("SansSerif", Font.BOLD, 20));
        bookNowBtn.setBackground(new Color(202, 97, 61));
        bookNowBtn.setForeground(Color.WHITE);
        bookNowBtn.setBounds(contentLeft, 320, 220, 45);
        bookNowBtn.setFocusPainted(false);
        overlayPanel.add(bookNowBtn);

        mainPanel.add(overlayPanel, BorderLayout.CENTER);

        // ---- Navigation Actions ----
        homeBtn.addActionListener(e -> showHomePage());
        aboutBtn.addActionListener(e -> showAboutUsPage());
        bookBtn.addActionListener(e -> showRoomsPage());
        contactBtn.addActionListener(e -> showContactPage());
        signInBtn.addActionListener(e -> showSignInPage());
        bookNowBtn.addActionListener(e -> showRoomsPage());

        // ---- Finalize Frame ----
        frame.setContentPane(mainPanel);
        frame.revalidate();
        frame.repaint();
        frame.setVisible(true);
    }



    // --- Sign In Page ---
    private void showSignInPage() {
        frame.getContentPane().removeAll();
```

```
// Background Panel
JPanel panel = new JPanel(null);
panel.setBackground(new Color(252, 250, 246));

// Company Title
JLabel title = new JLabel("StayZy");
title.setFont(new Font("Georgia", Font.BOLD, 42));
title.setForeground(new Color(202, 97, 61));
title.setBounds(330, 40, 400, 50);
panel.add(title);

// Subtitle
JLabel label = new JLabel("Sign In to Continue Your Journey");
label.setFont(new Font("Serif", Font.ITALIC, 18));
label.setForeground(new Color(80, 80, 80));
label.setBounds(260, 100, 400, 30);
panel.add(label);

// Username
JLabel userLabel = new JLabel("Username:");
userLabel.setFont(new Font("SansSerif", Font.PLAIN, 16));
userLabel.setBounds(250, 170, 100, 25);
panel.add(userLabel);

JTextField userField = new JTextField();
userField.setBounds(360, 170, 200, 30);
userField.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200), 1, true));
panel.add(userField);

// Email
JLabel emailLabel = new JLabel("Email:");
emailLabel.setFont(new Font("SansSerif", Font.PLAIN, 16));
emailLabel.setBounds(250, 220, 100, 25);
panel.add(emailLabel);

JTextField emailField = new JTextField();
emailField.setBounds(360, 220, 200, 30);
emailField.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200), 1, true));
panel.add(emailField);
```

```
// Password
JLabel passLabel = new JLabel("Password:");
passLabel.setFont(new Font("SansSerif", Font.PLAIN, 16));
passLabel.setBounds(250, 270, 100, 25);
panel.add(passLabel);

JPasswordField passField = new JPasswordField();
passField.setBounds(360, 270, 200, 30);
passField.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200), 1, true));
panel.add(passField);

// Login Button
JButton loginBtn = new JButton("Login");
loginBtn.setBounds(330, 330, 120, 35);
loginBtn.setBackground(new Color(202, 97, 61));
loginBtn.setForeground(Color.WHITE);
loginBtn.setFocusPainted(false);
loginBtn.setFont(new Font("SansSerif", Font.BOLD, 16));
loginBtn.setBorder(BorderFactory.createEmptyBorder());
panel.add(loginBtn);

// Back Button
JButton backBtn = new JButton("Back");
backBtn.setBounds(470, 330, 90, 35);
backBtn.setBackground(new Color(230, 230, 230));
backBtn.setFont(new Font("SansSerif", Font.PLAIN, 14));
backBtn.addActionListener(e -> showHomePage());
panel.add(backBtn);

// Action for login button
loginBtn.addActionListener(e -> {
    String username = userField.getText().trim();
    String email = emailField.getText().trim();
    String password = new String(passField.getPassword());

    if (username.isEmpty() || email.isEmpty() || password.isEmpty()) {
        JOptionPane.showMessageDialog(frame, "Please fill all fields.", "Error", JOptionPane.ERROR_MESSAGE);
    } else {
        showWelcomePage(username);
    }
});
```

```
    frame.add(panel);
    frame.revalidate();
    frame.repaint();
}

// --- Welcome Page ---
private void showWelcomePage(String username) {
    frame.getContentPane().removeAll();

    JPanel panel = new JPanel(null);
    panel.setBackground(new Color(252, 250, 246));

    JLabel welcome = new JLabel("Welcome to StayZy, " + username + "!");
    welcome.setFont(new Font("Georgia", Font.BOLD, 28));
    welcome.setForeground(new Color(202, 97, 61));
    welcome.setBounds(200, 200, 500, 40);
    panel.add(welcome);

    JButton homeBtn = new JButton("Go to Home");
    homeBtn.setBounds(320, 300, 150, 35);
    homeBtn.setBackground(new Color(202, 97, 61));
    homeBtn.setForeground(Color.WHITE);
    homeBtn.setFocusPainted(false);
    homeBtn.addActionListener(e -> showHomePage());
    panel.add(homeBtn);

    frame.add(panel);
    frame.revalidate();
    frame.repaint();
}

private void showRoomsPage() {
    frame.getContentPane().removeAll();

    JPanel mainPanel = new JPanel(new BorderLayout());
    mainPanel.setBackground(new Color(252, 250, 246));
    mainPanel.setBorder(new EmptyBorder(30, 60, 30, 60));

    // --- Card Panel for Room Search ---
    JPanel cardPanel = new JPanel();
```

```
cardPanel.setLayout(new BoxLayout(cardPanel, BoxLayout.Y_AXIS));
cardPanel.setBackground(Color.WHITE);
cardPanel.setBorder(new CompoundBorder(
   new LineBorder(new Color(202, 97, 61), 2, true),
   new EmptyBorder(30, 60, 30, 60)
));

JLabel title = new JLabel("Book Your Room");
title.setFont(new Font("Georgia", Font.BOLD, 32));
title.setForeground(new Color(202, 97, 61));
title.setAlignmentX(Component.CENTER_ALIGNMENT);
cardPanel.add(title);
cardPanel.add(Box.createVerticalStrut(20));

// Destination Dropdown (non-editable, loads all locations)
JPanel destPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
destPanel.setOpaque(false);
JLabel lblDestination = new JLabel("Destination:");
lblDestination.setFont(new Font("SansSerif", Font.PLAIN, 16));
JComboBox<String> cbDestination = new JComboBox<>();
cbDestination.setFont(new Font("SansSerif", Font.PLAIN, 16));
cbDestination.setEditable(false); // Non-editable dropdown
cbDestination.addItem("Select"); // Show 'Select' while loading
for (String loc : fetchAllLocations()) {
   cbDestination.addItem(loc);
}
destPanel.add(lblDestination);
destPanel.add(cbDestination);
cardPanel.add(destPanel);
cardPanel.add(Box.createVerticalStrut(10));

// Check-in Date (Calendar)
JPanel checkInPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
checkInPanel.setOpaque(false);
JLabel lblCheckIn = new JLabel("Check-in Date:");
lblCheckIn.setFont(new Font("SansSerif", Font.PLAIN, 16));
UtilDateModel checkInModel = new UtilDateModel();
Properties p = new Properties();
p.put("text.today", "Today");
p.put("text.month", "Month");
p.put("text.year", "Year");
```

```
JDatePanelImpl checkInDatePanel = new JDatePanelImpl(checkInModel, p);
JDatePickerImpl checkInPicker = new JDatePickerImpl(checkInDatePanel, new DateLabelFormatter());
checkInPanel.add(lblCheckIn);
checkInPanel.add(checkInPicker);
cardPanel.add(checkInPanel);
cardPanel.add(Box.createVerticalStrut(10));

// Check-out Date (Calendar)
JPanel checkOutPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
checkOutPanel.setOpaque(false);
JLabel lblCheckOut = new JLabel("Check-out Date:");
lblCheckOut.setFont(new Font("SansSerif", Font.PLAIN, 16));
UtilDateModel checkOutModel = new UtilDateModel();
JDatePanelImpl checkOutDatePanel = new JDatePanelImpl(checkOutModel, p);
JDatePickerImpl checkOutPicker = new JDatePickerImpl(checkOutDatePanel, new DateLabelFormatter());
checkOutPanel.add(lblCheckOut);
checkOutPanel.add(checkOutPicker);
cardPanel.add(checkOutPanel);
cardPanel.add(Box.createVerticalStrut(10));

// Number of Guests
JPanel guestsPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
guestsPanel.setOpaque(false);
JLabel lblGuests = new JLabel("Number of Guests:");
lblGuests.setFont(new Font("SansSerif", Font.PLAIN, 16));
JComboBox<String> cbGuests = new JComboBox<>(new String[]{"1", "2", "3", "4", "5+"});
cbGuests.setFont(new Font("SansSerif", Font.PLAIN, 16));
guestsPanel.add(lblGuests);
guestsPanel.add(cbGuests);
cardPanel.add(guestsPanel);
cardPanel.add(Box.createVerticalStrut(20));

// Search Button (was Enter)
JButton btnSearch = new JButton("Search");
btnSearch.setBackground(new Color(202, 97, 61));
btnSearch.setForeground(Color.WHITE);
btnSearch.setFont(new Font("SansSerif", Font.BOLD, 16));
btnSearch.setFocusPainted(false);
btnSearch.setAlignmentX(Component.CENTER_ALIGNMENT);
cardPanel.add(btnSearch);
cardPanel.add(Box.createVerticalStrut(10));
```

```
        // Back Button
        JButton btnBack = new JButton("Back");
        btnBack.setBackground(new Color(230, 230, 230));
        btnBack.setFont(new Font("SansSerif", Font.PLAIN, 14));
        btnBack.setAlignmentX(Component.CENTER_ALIGNMENT);
        btnBack.addActionListener(e -> showHomePage());
        cardPanel.add(btnBack);

        // Action for Search button
        btnSearch.addActionListener(e -> {
            String destination = (String) cbDestination.getSelectedItem();
            // Prevent search if 'Select' is chosen
            if (destination == null || destination.equals("Select")) {
                JOptionPane.showMessageDialog(frame, "Please select a destination.", "Error",
JOptionPane.ERROR_MESSAGE);
                return;
            }
            showResultsPage(destination, "rooms");
        });

        mainPanel.add(cardPanel, BorderLayout.CENTER);
        frame.add(mainPanel);
        frame.revalidate();
        frame.repaint();
    }

    // --- Results Page ---
    private void showResultsPage(String destination, String fromPage) {
        frame.getContentPane().removeAll();
        JPanel mainPanel = new JPanel(new BorderLayout());
        mainPanel.setBackground(new Color(252, 250, 246));

        JLabel label = new JLabel("Hotels in " + destination);
        label.setFont(new Font("Georgia", Font.BOLD, 32));
        label.setForeground(new Color(202, 97, 61));
        label.setBorder(new EmptyBorder(20, 0, 20, 0));
        label.setHorizontalAlignment(SwingConstants.CENTER);
        mainPanel.add(label, BorderLayout.NORTH);

        // Fetch hotels from DB for the selected location
```

```
ArrayList<String> hotelsList = fetchHotelsByLocation(destination);
//System.out.println("[DEBUG] Hotels found for location '" + destination + "': " + hotelsList.size());

JPanel gridPanel = new JPanel();
gridPanel.setLayout(new BoxLayout(gridPanel, BoxLayout.Y_AXIS)); // One hotel per row
gridPanel.setBackground(new Color(252, 250, 246));
gridPanel.setBorder(new EmptyBorder(20, 40, 20, 40));

if (hotelsList.isEmpty()) {
    JLabel noHotels = new JLabel("No hotels found for this location.");
    noHotels.setFont(new Font("Serif", Font.BOLD, 20));
    noHotels.setForeground(new Color(180, 50, 50));
    noHotels.setHorizontalAlignment(SwingConstants.CENTER);
    gridPanel.add(noHotels);
} else {
    for (String hotel : hotelsList) {
        int hotelId = fetchHotelIdByName(hotel);
        String amenities = fetchHotelAmenitiesById(hotelId);
        ArrayList<String[]> rooms = fetchRoomsByHotelId(hotelId);
        //System.out.println("[DEBUG] Hotel: " + hotel + " (ID: " + hotelId + ") Rooms found: " + rooms.size());
        // Only show available rooms
        ArrayList<String[]> availableRooms = new ArrayList<>();
        for (String[] r : rooms) {
            if (r[2] != null && r[2].equalsIgnoreCase("available")) {
                availableRooms.add(r);
            }
        }
        //System.out.println("[DEBUG] Hotel: " + hotel + " Available rooms: " + availableRooms.size());
        if (availableRooms.isEmpty()) continue;

        JPanel card = new JPanel();
        card.setLayout(new BoxLayout(card, BoxLayout.Y_AXIS));
        card.setBackground(Color.WHITE);
        card.setBorder(new CompoundBorder(
            new LineBorder(new Color(202, 97, 61), 2, true),
            new EmptyBorder(20, 20, 20, 20)
        ));
        card.setMaximumSize(new Dimension(Integer.MAX_VALUE, 220));

        // Top: Hotel name (left aligned)
        JPanel topPanel = new JPanel(new FlowLayout(FlowLayout.LEFT, 0, 0));
```

```
topPanel.setOpaque(false);
JLabel nameLabel = new JLabel(hotel);
nameLabel.setFont(new Font("Georgia", Font.BOLD, 22));
nameLabel.setForeground(new Color(60, 60, 60));
topPanel.add(nameLabel);
card.add(topPanel);
// Amenities directly below hotel name, left aligned
JPanel amenitiesPanel = new JPanel(new FlowLayout(FlowLayout.LEFT, 0, 0));
amenitiesPanel.setOpaque(false);
JLabel amenitiesLabel = new JLabel("<html><b>Amenities:</b> " + amenities + "</html>");
amenitiesLabel.setFont(new Font("SansSerif", Font.PLAIN, 14));
amenitiesLabel.setForeground(new Color(90, 90, 90));
amenitiesLabel.setBorder(new EmptyBorder(4, 0, 10, 0));
amenitiesPanel.add(amenitiesLabel);
card.add(amenitiesPanel);

// For each available room, show as a row
for (String[] row : availableRooms) {
    JPanel roomPanel = new JPanel(new BorderLayout());
    roomPanel.setPreferredSize(new Dimension(0, 48));
    roomPanel.setMaximumSize(new Dimension(Integer.MAX_VALUE, 48));
    roomPanel.setBackground(new Color(250, 245, 240));
    roomPanel.setBorder(new CompoundBorder(
        new MatteBorder(0, 0, 1, 0, new Color(230, 230, 230)),
        new EmptyBorder(0, 0, 0, 0)));

    // Room Type (left)
    JLabel roomType = new JLabel(row[0]);
    roomType.setFont(new Font("SansSerif", Font.BOLD, 15));
    roomType.setForeground(new Color(60, 60, 60));
    roomType.setBorder(new EmptyBorder(0, 16, 0, 0));
    roomPanel.add(roomType, BorderLayout.WEST);

    // Price (right)
    JLabel price = new JLabel("₹" + row[1]);
    price.setFont(new Font("SansSerif", Font.BOLD, 16));
    price.setForeground(new Color(202, 97, 61));
    price.setHorizontalAlignment(SwingConstants.RIGHT);
    price.setBorder(new EmptyBorder(0, 0, 0, 16));
    roomPanel.add(price, BorderLayout.EAST);
```

```
            // Select Button (right, not overlapping)
            JButton selectBtn = new JButton("Select");
            selectBtn.setBackground(new Color(202, 97, 61));
            selectBtn.setForeground(Color.WHITE);
            selectBtn.setFont(new Font("SansSerif", Font.BOLD, 14));
            selectBtn.setFocusPainted(false);
            selectBtn.setPreferredSize(new Dimension(90, 32));
            JPanel btnPanel = new JPanel(new FlowLayout(FlowLayout.RIGHT, 0, 0));
            btnPanel.setOpaque(false);
            btnPanel.add(selectBtn);
            roomPanel.add(btnPanel, BorderLayout.CENTER);
            selectBtn.addActionListener(e -> showLoginPopup(hotel, row));

            card.add(roomPanel);
        }
        gridPanel.add(Box.createVerticalStrut(10));
        gridPanel.add(card);
    }
}

JScrollPane scrollPane = new JScrollPane(gridPanel);
scrollPane.setHorizontalScrollBarPolicy(ScrollPaneConstants.HORIZONTAL_SCROLLBAR_NEVER);
scrollPane.setVerticalScrollBarPolicy(ScrollPaneConstants.VERTICAL_SCROLLBAR_NEVER);
scrollPane.setPreferredSize(new Dimension(frame.getWidth(), frame.getHeight() - 100));
mainPanel.add(scrollPane, BorderLayout.CENTER);
frame.setExtendedState(JFrame.MAXIMIZED_BOTH);

// Back Button
JPanel bottomPanel = new JPanel();
bottomPanel.setBackground(new Color(252, 250, 246));
JButton backBtn = new JButton("Back");
backBtn.setBackground(new Color(202, 97, 61));
backBtn.setForeground(Color.WHITE);
backBtn.setFont(new Font("SansSerif", Font.BOLD, 15));
backBtn.setFocusPainted(false);
backBtn.addActionListener(e -> {
    if ("rooms".equals(fromPage)) {
        showRoomsPage();
    } else {
        showSearchPage();
    }
```

```
        });
        bottomPanel.add(backBtn);
        mainPanel.add(bottomPanel, BorderLayout.SOUTH);

        frame.add(mainPanel);
        frame.revalidate();
        frame.repaint();
    }

    // Show login popup when Select is clicked
    private void showLoginPopup(String hotel, String[] room) {
        JDialog dialog = new JDialog(frame, "Login to Book", true);
        dialog.setSize(400, 320);
        dialog.setLocationRelativeTo(frame);
        dialog.setLayout(null);
        dialog.getContentPane().setBackground(new Color(252, 250, 246));

        JLabel title = new JLabel("Sign In to Continue");
        title.setFont(new Font("Georgia", Font.BOLD, 20));
        title.setForeground(new Color(202, 97, 61));
        title.setBounds(90, 20, 250, 30);
        dialog.add(title);

        JLabel userLabel = new JLabel("Email:");
        userLabel.setFont(new Font("SansSerif", Font.PLAIN, 15));
        userLabel.setBounds(50, 70, 80, 25);
        dialog.add(userLabel);
        JTextField userField = new JTextField();
        userField.setBounds(140, 70, 180, 28);
        dialog.add(userField);

        JLabel passLabel = new JLabel("Password:");
        passLabel.setFont(new Font("SansSerif", Font.PLAIN, 15));
        passLabel.setBounds(50, 110, 80, 25);
        dialog.add(passLabel);
        JPasswordField passField = new JPasswordField();
        passField.setBounds(140, 110, 180, 28);
        dialog.add(passField);

        JButton loginBtn = new JButton("Login");
        loginBtn.setBackground(new Color(202, 97, 61));
```

```
        loginBtn.setForeground(Color.WHITE);
        loginBtn.setFont(new Font("SansSerif", Font.BOLD, 15));
        loginBtn.setBounds(140, 160, 100, 32);
        dialog.add(loginBtn);

        JLabel registerLabel = new JLabel("New user? Register here");
        registerLabel.setFont(new Font("SansSerif", Font.PLAIN, 13));
        registerLabel.setForeground(new Color(60, 60, 200));
        registerLabel.setBounds(120, 210, 180, 22);
        registerLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
        dialog.add(registerLabel);

        // Register link action
        registerLabel.addMouseListener(new java.awt.event.MouseAdapter() {
            public void mouseClicked(java.awt.event.MouseEvent evt) {
                dialog.dispose();
                showRegisterPopup(hotel, room);
            }
        });

        // Login button action
        loginBtn.addActionListener(e -> {
            String email = userField.getText().trim();
            String password = new String(passField.getPassword());
            if (email.isEmpty() || password.isEmpty()) {
                JOptionPane.showMessageDialog(dialog, "Please enter email and password.", "Error",
JOptionPane.ERROR_MESSAGE);
                return;
            }
            if (validateUserLogin(email, password)) {
                dialog.dispose();
                showGuestDetailsPage(email, hotel, room);
            } else {
                JOptionPane.showMessageDialog(dialog, "Invalid credentials.", "Login Failed",
JOptionPane.ERROR_MESSAGE);
            }
        });

        dialog.setVisible(true);
    }
```

```
// Validate user login (SHA-256 hash check)
private boolean validateUserLogin(String email, String password) {
    String query = "SELECT password FROM userdetails WHERE email = ?";
    try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
         PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setString(1, email);
        ResultSet rs = stmt.executeQuery();
        if (rs.next()) {
            String dbHash = rs.getString("password");
            String inputHash = sha256(password);
            return dbHash != null && dbHash.equalsIgnoreCase(inputHash);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return false;
}

// SHA-256 hash utility
private String sha256(String base) {
    try {
        java.security.MessageDigest digest = java.security.MessageDigest.getInstance("SHA-256");
        byte[] hash = digest.digest(base.getBytes("UTF-8"));
        StringBuilder hexString = new StringBuilder();
        for (byte b : hash) {
            String hex = Integer.toHexString(0xff & b);
            if(hex.length() == 1) hexString.append('0');
            hexString.append(hex);
        }
        return hexString.toString();
    } catch(Exception ex){
        throw new RuntimeException(ex);
    }
}

// Show register popup
private void showRegisterPopup(String hotel, String[] room) {
    JDialog dialog = new JDialog(frame, "Register New User", true);
    dialog.setSize(420, 550);
    dialog.setLocationRelativeTo(frame);
    dialog.setLayout(null);
```

```
dialog.getContentPane().setBackground(new Color(252, 250, 246));

JLabel title = new JLabel("Register");
title.setFont(new Font("Georgia", Font.BOLD, 20));
title.setForeground(new Color(202, 97, 61));
title.setBounds(150, 18, 200, 30);
dialog.add(title);

JLabel fnameLabel = new JLabel("First Name:");
fnameLabel.setBounds(40, 60, 100, 25);
dialog.add(fnameLabel);
JTextField fnameField = new JTextField();
fnameField.setBounds(150, 60, 200, 25);
dialog.add(fnameField);

JLabel lnameLabel = new JLabel("Last Name:");
lnameLabel.setBounds(40, 95, 100, 25);
dialog.add(lnameLabel);
JTextField lnameField = new JTextField();
lnameField.setBounds(150, 95, 200, 25);
dialog.add(lnameField);

JLabel emailLabel = new JLabel("Email:");
emailLabel.setBounds(40, 130, 100, 25);
dialog.add(emailLabel);
JTextField emailField = new JTextField();
emailField.setBounds(150, 130, 200, 25);
dialog.add(emailField);

JLabel passLabel = new JLabel("Password:");
passLabel.setBounds(40, 165, 100, 25);
dialog.add(passLabel);
JPasswordField passField = new JPasswordField();
passField.setBounds(150, 165, 200, 25);
dialog.add(passField);

JLabel phoneLabel = new JLabel("Phone:");
phoneLabel.setBounds(40, 200, 100, 25);
dialog.add(phoneLabel);
JTextField phoneField = new JTextField();
phoneField.setBounds(150, 200, 200, 25);
```

```
dialog.add(phoneField);

JLabel altPhoneLabel = new JLabel("Alt. Phone:");
altPhoneLabel.setBounds(40, 235, 100, 25);
dialog.add(altPhoneLabel);
JTextField altPhoneField = new JTextField();
altPhoneField.setBounds(150, 235, 200, 25);
dialog.add(altPhoneField);

JLabel dobLabel = new JLabel("DOB:");
dobLabel.setBounds(40, 270, 100, 25);
dialog.add(dobLabel);
// Use JDatePicker for DOB
UtilDateModel dobModel = new UtilDateModel();
Properties p = new Properties();
p.put("text.today", "Today");
p.put("text.month", "Month");
p.put("text.year", "Year");
JDatePanelImpl dobDatePanel = new JDatePanelImpl(dobModel, p);
JDatePickerImpl dobPicker = new JDatePickerImpl(dobDatePanel, new DateLabelFormatter());
dobPicker.setBounds(150, 270, 200, 30);
dialog.add(dobPicker);

JLabel addressLabel = new JLabel("Address:");
addressLabel.setBounds(40, 310, 100, 25);
dialog.add(addressLabel);
JTextField addressField = new JTextField();
addressField.setBounds(150, 310, 200, 25);
dialog.add(addressField);

JLabel cityLabel = new JLabel("City:");
cityLabel.setBounds(40, 345, 100, 25);
dialog.add(cityLabel);
JTextField cityField = new JTextField();
cityField.setBounds(150, 345, 200, 25);
dialog.add(cityField);

JLabel stateLabel = new JLabel("State:");
stateLabel.setBounds(40, 380, 100, 25);
dialog.add(stateLabel);
JTextField stateField = new JTextField();
```

```
        stateField.setBounds(150, 380, 200, 25);
        dialog.add(stateField);

        JLabel countryLabel = new JLabel("Country:");
        countryLabel.setBounds(40, 415, 100, 25);
        dialog.add(countryLabel);
        JTextField countryField = new JTextField();
        countryField.setBounds(150, 415, 200, 25);
        dialog.add(countryField);

        JButton regBtn = new JButton("Register");
        regBtn.setBackground(new Color(202, 97, 61));
        regBtn.setForeground(Color.WHITE);
        regBtn.setFont(new Font("SansSerif", Font.BOLD, 15));
        regBtn.setBounds(150, 460, 120, 32);
        dialog.add(regBtn);
        dialog.setSize(420, 550); // Increase popup height

        regBtn.addActionListener(e -> {
            String fname = fnameField.getText().trim();
            String lname = lnameField.getText().trim();
            String email = emailField.getText().trim();
            String pass = new String(passField.getPassword());
            String phone = phoneField.getText().trim();
            String altPhone = altPhoneField.getText().trim();
            // Get DOB from date picker
            java.util.Date dobDate = (java.util.Date) dobPicker.getModel().getValue();
            String dobFormatted = "";
            if (dobDate != null) {
                java.text.SimpleDateFormat outFmt = new java.text.SimpleDateFormat("yyyy-MM-dd");
                dobFormatted = outFmt.format(dobDate);
            }
            String address = addressField.getText().trim();
            String city = cityField.getText().trim();
            String state = stateField.getText().trim();
            String country = countryField.getText().trim();
            if (fname.isEmpty() || lname.isEmpty() || email.isEmpty() || pass.isEmpty() || phone.isEmpty() ||
dobFormatted.isEmpty() || address.isEmpty() || city.isEmpty() || state.isEmpty() || country.isEmpty()) {
                JOptionPane.showMessageDialog(dialog, "Please fill all fields.", "Error", JOptionPane.ERROR_MESSAGE);
                return;
            }
```

```
            if (registerUser(fname, lname, email, pass, phone, altPhone, dobFormatted, address, city, state, country)) {
                dialog.dispose();
                // Fetch amenities for this hotel
                int hotelId = fetchHotelIdByName(hotel);
                String amenities = fetchHotelAmenitiesById(hotelId);
                showPaymentPage(hotel, room, amenities, fname, lname, email);
            } else {
                JOptionPane.showMessageDialog(dialog, "Registration failed. Email may already exist.", "Error",
JOptionPane.ERROR_MESSAGE);
            }
        });

        dialog.setVisible(true);
    }

    // Show payment page after registration
    private void showPaymentPage(String hotel, String[] room, String amenities, String fname, String lname, String email) {
        frame.getContentPane().removeAll();
        JPanel mainPanel = new JPanel();
        mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.Y_AXIS));
        mainPanel.setBackground(new Color(252, 250, 246));
        mainPanel.setBorder(new EmptyBorder(30, 60, 30, 60));

        // Top: Selected hotel, room, amenities
        JPanel infoPanel = new JPanel();
        infoPanel.setLayout(new BoxLayout(infoPanel, BoxLayout.Y_AXIS));
        infoPanel.setBackground(Color.WHITE);
        infoPanel.setBorder(new CompoundBorder(
            new LineBorder(new Color(202, 97, 61), 2, true),
            new EmptyBorder(20, 30, 20, 30)
        ));
        JLabel hotelLabel = new JLabel(hotel);
        hotelLabel.setFont(new Font("Georgia", Font.BOLD, 22));
        hotelLabel.setForeground(new Color(60, 60, 60));
        infoPanel.add(hotelLabel);
        infoPanel.add(Box.createVerticalStrut(8));
        JLabel roomLabel = new JLabel("Room: " + room[0] + "   Price: ₹" + room[1]);
        roomLabel.setFont(new Font("SansSerif", Font.BOLD, 16));
        roomLabel.setForeground(new Color(202, 97, 61));
        infoPanel.add(roomLabel);
        infoPanel.add(Box.createVerticalStrut(6));
```

```
JLabel amenitiesLabel = new JLabel("<html><b>Amenities:</b> " + amenities + "</html>");
amenitiesLabel.setFont(new Font("SansSerif", Font.PLAIN, 14));
amenitiesLabel.setForeground(new Color(90, 90, 90));
infoPanel.add(amenitiesLabel);
mainPanel.add(infoPanel);
mainPanel.add(Box.createVerticalStrut(30));

// Payment form (simple demo)
JPanel payPanel = new JPanel(new GridBagLayout());
payPanel.setBackground(Color.WHITE);
payPanel.setBorder(new CompoundBorder(
   new LineBorder(new Color(202, 97, 61), 1, true),
   new EmptyBorder(20, 30, 20, 30)
));
GridBagConstraints gbc = new GridBagConstraints();
gbc.insets = new Insets(8, 8, 8, 8);
gbc.anchor = GridBagConstraints.WEST;
gbc.fill = GridBagConstraints.HORIZONTAL;
gbc.gridx = 0;
gbc.gridy = 0;
gbc.gridwidth = 2;
JLabel payTitle = new JLabel("Payment Details");
payTitle.setFont(new Font("Georgia", Font.BOLD, 18));
payTitle.setForeground(new Color(202, 97, 61));
payTitle.setHorizontalAlignment(SwingConstants.CENTER);
payPanel.add(payTitle, gbc);
gbc.gridy++;
gbc.gridwidth = 1;
payPanel.add(new JLabel("Name:"), gbc);
gbc.gridx = 1;
JLabel nameLabel = new JLabel(fname + " " + lname);
nameLabel.setFont(new Font("SansSerif", Font.PLAIN, 15));
payPanel.add(nameLabel, gbc);
gbc.gridx = 0;
gbc.gridy++;
payPanel.add(new JLabel("Email:"), gbc);
gbc.gridx = 1;
JLabel emailLabel2 = new JLabel(email);
emailLabel2.setFont(new Font("SansSerif", Font.PLAIN, 15));
payPanel.add(emailLabel2, gbc);
gbc.gridx = 0;
```

```
        gbc.gridy++;
        payPanel.add(new JLabel("Card Number:"), gbc);
        gbc.gridx = 1;
        JTextField cardField = new JTextField();
        cardField.setFont(new Font("SansSerif", Font.PLAIN, 15));
        cardField.setDocument(new javax.swing.text.PlainDocument() {
            @Override
            public void insertString(int offs, String str, javax.swing.text.AttributeSet a) throws
javax.swing.text.BadLocationException {
                if (str == null) return;
                if ((getLength() + str.length()) <= 16 && str.matches("\\d+")) {
                    super.insertString(offs, str, a);
                }
            }
        });
        payPanel.add(cardField, gbc);
        gbc.gridx = 0;
        gbc.gridy++;
        payPanel.add(new JLabel("CVV:"), gbc);
        gbc.gridx = 1;
        JTextField cvvField = new JTextField();
        cvvField.setFont(new Font("SansSerif", Font.PLAIN, 15));
        cvvField.setDocument(new javax.swing.text.PlainDocument() {
            @Override
            public void insertString(int offs, String str, javax.swing.text.AttributeSet a) throws
javax.swing.text.BadLocationException {
                if (str == null) return;
                if ((getLength() + str.length()) <= 4 && str.matches("\\d+")) {
                    super.insertString(offs, str, a);
                }
            }
        });
        payPanel.add(cvvField, gbc);
        gbc.gridx = 0;
        gbc.gridy++;
        payPanel.add(new JLabel("Expiry (MM/YY):"), gbc);
        gbc.gridx = 1;
        JTextField expField = new JTextField();
        expField.setFont(new Font("SansSerif", Font.PLAIN, 15));
        payPanel.add(expField, gbc);
        gbc.gridx = 0;
```

```
        gbc.gridy++;
        gbc.gridwidth = 2;
        gbc.anchor = GridBagConstraints.CENTER;
        JButton payBtn = new JButton("Pay Now");
        payBtn.setBackground(new Color(202, 97, 61));
        payBtn.setForeground(Color.WHITE);
        payBtn.setFont(new Font("SansSerif", Font.BOLD, 15));
        payPanel.add(payBtn, gbc);
        mainPanel.add(payPanel);

        // Pay button action (demo)
        payBtn.addActionListener(e -> {
            // Generate a random confirmation/booking ID
            int bookingId = (int) (Math.random() * 900000) + 100000;
            // For demo, use today's date as check-in and tomorrow as check-out
            java.text.SimpleDateFormat sdf = new java.text.SimpleDateFormat("yyyy-MM-dd");
            String checkIn = sdf.format(new java.util.Date());
            String checkOut = sdf.format(new java.util.Date(System.currentTimeMillis() + 24*60*60*1000));
            int numPassengers = 1; // You can pass this from previous page if needed
            showConfirmationPage(bookingId, fname + " " + lname, email, room[0], checkIn, checkOut, hotel, room[1],
amenities, numPassengers);
        });

        frame.add(mainPanel);
        frame.revalidate();
        frame.repaint();
    }

    // Register user (minimal fields for demo)
    private boolean registerUser(String fname, String lname, String email, String password, String phone, String altPhone,
String dob, String address, String city, String state, String country) {
        String query = "INSERT INTO userdetails (firstname, lastname, email, password, phone, alternate_phone, dob, address,
city, state, country) VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             PreparedStatement stmt = conn.prepareStatement(query)) {
            System.out.println("[DEBUG] SQL: " + query);
            System.out.println("[DEBUG] Params: " + fname + ", " + lname + ", " + email + ", " + sha256(password) + ", " +
phone + ", " + altPhone + ", " + dob + ", " + address + ", " + city + ", " + state + ", " + country);
            stmt.setString(1, fname);
            stmt.setString(2, lname);
            stmt.setString(3, email);
```

```
            stmt.setString(4, sha256(password));
            stmt.setString(5, phone);
            stmt.setString(6, altPhone);
            stmt.setString(7, dob);
            stmt.setString(8, address);
            stmt.setString(9, city);
            stmt.setString(10, state);
            stmt.setString(11, country);
            stmt.executeUpdate();
            return true;
        } catch (Exception e) {
            e.printStackTrace();
        }
        return false;
    }

    // Show guest details page after login/register
    private void showGuestDetailsPage(String email, String hotel, String[] room) {
        // Fetch user details from DB
        String fname = "", lname = "", phone = "", altPhone = "", dob = "", address = "", city = "", state = "", country = "",
userEmail = email;
        String query = "SELECT * FROM userdetails WHERE email = ?";
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             PreparedStatement stmt = conn.prepareStatement(query)) {
            stmt.setString(1, email);
            ResultSet rs = stmt.executeQuery();
            if (rs.next()) {
                fname = rs.getString("firstname");
                lname = rs.getString("lastname");
                phone = rs.getString("phone");
                altPhone = rs.getString("alternate_phone");
                dob = rs.getString("dob");
                address = rs.getString("address");
                city = rs.getString("city");
                state = rs.getString("state");
                country = rs.getString("country");
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        // Show guest details form
```

```
JDialog dialog = new JDialog(frame, "Guest Details", true);
dialog.setSize(500, 500);
dialog.setLocationRelativeTo(frame);
dialog.setLayout(null);
dialog.getContentPane().setBackground(new Color(252, 250, 246));

JLabel title = new JLabel("Guest Details for " + hotel + " - " + room[0]);
title.setFont(new Font("Georgia", Font.BOLD, 18));
title.setForeground(new Color(202, 97, 61));
title.setBounds(60, 18, 400, 30);
dialog.add(title);

JLabel fnameLabel = new JLabel("First Name:");
fnameLabel.setBounds(40, 60, 100, 25);
dialog.add(fnameLabel);
JTextField fnameField = new JTextField(fname);
fnameField.setBounds(150, 60, 250, 25);
dialog.add(fnameField);

JLabel lnameLabel = new JLabel("Last Name:");
lnameLabel.setBounds(40, 95, 100, 25);
dialog.add(lnameLabel);
JTextField lnameField = new JTextField(lname);
lnameField.setBounds(150, 95, 250, 25);
dialog.add(lnameField);

JLabel emailLabel = new JLabel("Email:");
emailLabel.setBounds(40, 130, 100, 25);
dialog.add(emailLabel);
JTextField emailField = new JTextField(userEmail);
emailField.setBounds(150, 130, 250, 25);
dialog.add(emailField);

JLabel phoneLabel = new JLabel("Phone:");
phoneLabel.setBounds(40, 165, 100, 25);
dialog.add(phoneLabel);
JTextField phoneField = new JTextField(phone);
phoneField.setBounds(150, 165, 250, 25);
dialog.add(phoneField);

JLabel altPhoneLabel = new JLabel("Alt. Phone:");
```

```
altPhoneLabel.setBounds(40, 200, 100, 25);
dialog.add(altPhoneLabel);
JTextField altPhoneField = new JTextField(altPhone);
altPhoneField.setBounds(150, 200, 250, 25);
dialog.add(altPhoneField);

JLabel dobLabel = new JLabel("DOB:");
dobLabel.setBounds(40, 235, 100, 25);
dialog.add(dobLabel);
JTextField dobField = new JTextField(dob);
dobField.setBounds(150, 235, 250, 25);
dialog.add(dobField);

JLabel addressLabel = new JLabel("Address:");
addressLabel.setBounds(40, 270, 100, 25);
dialog.add(addressLabel);
JTextField addressField = new JTextField(address);
addressField.setBounds(150, 270, 250, 25);
dialog.add(addressField);

JLabel cityLabel = new JLabel("City:");
cityLabel.setBounds(40, 305, 100, 25);
dialog.add(cityLabel);
JTextField cityField = new JTextField(city);
cityField.setBounds(150, 305, 250, 25);
dialog.add(cityField);

JLabel stateLabel = new JLabel("State:");
stateLabel.setBounds(40, 340, 100, 25);
dialog.add(stateLabel);
JTextField stateField = new JTextField(state);
stateField.setBounds(150, 340, 250, 25);
dialog.add(stateField);

JLabel countryLabel = new JLabel("Country:");
countryLabel.setBounds(40, 375, 100, 25);
dialog.add(countryLabel);
JTextField countryField = new JTextField(country);
countryField.setBounds(150, 375, 250, 25);
dialog.add(countryField);
```

```
    JButton nextBtn = new JButton("Next");
    nextBtn.setBackground(new Color(202, 97, 61));
    nextBtn.setForeground(Color.WHITE);
    nextBtn.setFont(new Font("SansSerif", Font.BOLD, 15));
    nextBtn.setBounds(180, 420, 120, 32);
    dialog.add(nextBtn);

    // Next button action (proceed to booking/payment page)
    nextBtn.addActionListener(e -> {
        // Collect guest details (if you want to update DB, do it here)
        String updatedFname = fnameField.getText().trim();
        String updatedLname = lnameField.getText().trim();
        String updatedEmail = emailField.getText().trim();
        // Fetch amenities for this hotel
        int hotelId = fetchHotelIdByName(hotel);
        String amenities = fetchHotelAmenitiesById(hotelId);
        dialog.dispose();
        showPaymentPage(hotel, room, amenities, updatedFname, updatedLname, updatedEmail);
    });

    dialog.setVisible(true);
}

// Stub for showHotelDetailsPage
private void showHotelDetailsPage(String hotel, String fromPage) {
    JOptionPane.showMessageDialog(frame, "Hotel Details page for: " + hotel + " (from: " + fromPage + ")");
}

// Stub for showSearchPage
private void showSearchPage() {
    JOptionPane.showMessageDialog(frame, "Search page (not yet implemented)");
}

// Helper: Table header label
private JLabel tableHeaderLabel(String text) {
    JLabel l = new JLabel(text);
    l.setFont(new Font("SansSerif", Font.BOLD, 14));
    l.setForeground(new Color(202, 97, 61));
    l.setHorizontalAlignment(SwingConstants.CENTER);
    return l;
}
```

```
// Helper: Table cell label
private JLabel tableCellLabel(String text) {
    JLabel l = new JLabel(text);
    l.setFont(new Font("SansSerif", Font.PLAIN, 14));
    l.setForeground(new Color(60, 60, 60));
    l.setHorizontalAlignment(SwingConstants.CENTER);
    return l;
}

// Helper: Fetch hotelid by hotel name
private int fetchHotelIdByName(String hotelName) {
    int hotelId = -1;
    String query = "SELECT hotelid FROM hotels WHERE hotelname = ?";
    try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
         PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setString(1, hotelName);
        ResultSet rs = stmt.executeQuery();
        if (rs.next()) {
            hotelId = rs.getInt("hotelid");
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return hotelId;
}

// Helper: Fetch rooms by hotelid
private ArrayList<String[]> fetchRoomsByHotelId(int hotelId) {
    ArrayList<String[]> rooms = new ArrayList<>();
    String query = "SELECT room_type, price, status FROM rooms WHERE hotelid = ? ORDER BY room_type ASC";
    try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
         PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setInt(1, hotelId);
        ResultSet rs = stmt.executeQuery();
        while (rs.next()) {
            String[] room = new String[3];
            room[0] = rs.getString("room_type");
            room[1] = rs.getString("price");
            room[2] = rs.getString("status");
            rooms.add(room);
```

```
            }
            System.out.println("[DEBUG] fetchRoomsByHotelId for hotelId=" + hotelId + ": " + rooms.size() + " rooms");
            for (String[] r : rooms) {
                System.out.println("[DEBUG]   Room: type=" + r[0] + ", price=" + r[1] + ", status=" + r[2]);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return rooms;
    }

    // Helper: Fetch hotels by location name
    private ArrayList<String> fetchHotelsByLocation(String locationName) {
        ArrayList<String> hotels = new ArrayList<>();
        String query = "SELECT h.hotelname FROM hotels h JOIN location l ON h.locationid = l.locationid WHERE
l.locationname = ? ORDER BY h.hotelname ASC";
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             PreparedStatement stmt = conn.prepareStatement(query)) {
            stmt.setString(1, locationName);
            ResultSet rs = stmt.executeQuery();
            while (rs.next()) {
                hotels.add(rs.getString("hotelname"));
            }
            System.out.println("[DEBUG] fetchHotelsByLocation for '" + locationName + "': " + hotels);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return hotels;
    }

    // Helper: Fetch hotel amenities by hotel id
    private String fetchHotelAmenitiesById(int hotelId) {
        String amenities = "-";
        // Use correct table: htdetails (not hotels)
        String query = "SELECT amenities FROM htdetails WHERE hotelid = ?";
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             PreparedStatement stmt = conn.prepareStatement(query)) {
            stmt.setInt(1, hotelId);
            ResultSet rs = stmt.executeQuery();
            if (rs.next()) {
                amenities = rs.getString("amenities");
```

```
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return amenities;
    }

    // --- Confirmation Page ---
    private void showConfirmationPage(int bookingId, String customerName, String email, String roomType, String checkIn,
String checkOut, String hotelName, String price, String amenities, int numPassengers) {
        frame.getContentPane().removeAll();
        JPanel mainPanel = new JPanel(new BorderLayout());
        mainPanel.setBackground(new Color(252, 250, 246));

        // Success Icon and Message
        JPanel topPanel = new JPanel();
        topPanel.setOpaque(false);
        topPanel.setLayout(new BoxLayout(topPanel, BoxLayout.Y_AXIS));
        topPanel.setBorder(new EmptyBorder(40, 0, 10, 0));
        JLabel iconLabel = new JLabel("\u2714"); // Checkmark
        iconLabel.setFont(new Font("SansSerif", Font.BOLD, 64));
        iconLabel.setForeground(new Color(34, 139, 34));
        iconLabel.setAlignmentX(Component.CENTER_ALIGNMENT);
        topPanel.add(iconLabel);
        JLabel congrats = new JLabel("Booking Confirmed!");
        congrats.setFont(new Font("Georgia", Font.BOLD, 32));
        congrats.setForeground(new Color(202, 97, 61));
        congrats.setAlignmentX(Component.CENTER_ALIGNMENT);
        topPanel.add(congrats);
        JLabel thanks = new JLabel("Thank you for booking with StayZy, " + customerName + "!");
        thanks.setFont(new Font("SansSerif", Font.PLAIN, 18));
        thanks.setForeground(new Color(60, 60, 60));
        thanks.setAlignmentX(Component.CENTER_ALIGNMENT);
        topPanel.add(Box.createVerticalStrut(10));
        topPanel.add(thanks);
        mainPanel.add(topPanel, BorderLayout.NORTH);

        // Booking Details Card
        JPanel card = new JPanel();
        card.setLayout(new BoxLayout(card, BoxLayout.Y_AXIS));
        card.setBackground(Color.WHITE);
```

```
        card.setBorder(new CompoundBorder(
            new LineBorder(new Color(202, 97, 61), 2, true),
            new EmptyBorder(30, 40, 30, 40)
        ));
        card.setAlignmentX(Component.CENTER_ALIGNMENT);
        card.add(Box.createVerticalStrut(10));
        card.add(detailLabel("Confirmation ID:", String.valueOf(bookingId)));
        card.add(detailLabel("Hotel:", hotelName));
        card.add(detailLabel("Room Type:", roomType));
        card.add(detailLabel("Price:", "₹" + price));
        card.add(detailLabel("Check-in:", checkIn));
        card.add(detailLabel("Check-out:", checkOut));
        card.add(detailLabel("No. of Passengers:", String.valueOf(numPassengers)));
        card.add(detailLabel("Amenities:", amenities));
        card.add(detailLabel("Guest:", customerName));
        card.add(detailLabel("Email:", email));
        card.add(Box.createVerticalStrut(10));
        // Terms and Conditions
        JLabel terms = new JLabel("<html><b>Terms & Conditions:</b><br>• Please carry a valid ID proof at check-in.<br>• Free cancellation up to 24 hours before check-in.<br>• Check-in: 12:00 PM, Check-out: 11:00 AM.<br>• No smoking in rooms. Pets not allowed.<br>• For support, contact help@stayzy.com</html>");
        terms.setFont(new Font("SansSerif", Font.PLAIN, 13));
        terms.setForeground(new Color(90, 90, 90));
        card.add(terms);
        mainPanel.add(card, BorderLayout.CENTER);

        // Home Button
        JPanel bottomPanel = new JPanel();
        bottomPanel.setBackground(new Color(252, 250, 246));
        JButton homeBtn = new JButton("Back to Home");
        homeBtn.setBackground(new Color(202, 97, 61));
        homeBtn.setForeground(Color.WHITE);
        homeBtn.setFont(new Font("SansSerif", Font.BOLD, 16));
        homeBtn.setFocusPainted(false);
        homeBtn.addActionListener(e -> showHomePage());
        bottomPanel.add(homeBtn);
        mainPanel.add(bottomPanel, BorderLayout.SOUTH);

        frame.add(mainPanel);
        frame.revalidate();
        frame.repaint();
```

```
}

// Helper for booking detail rows
private JPanel detailLabel(String label, String value) {
    JPanel p = new JPanel(new FlowLayout(FlowLayout.LEFT));
    p.setOpaque(false);
    JLabel l = new JLabel(label);
    l.setFont(new Font("SansSerif", Font.BOLD, 15));
    l.setForeground(new Color(202, 97, 61));
    JLabel v = new JLabel(value);
    v.setFont(new Font("SansSerif", Font.PLAIN, 15));
    v.setForeground(new Color(60, 60, 60));
    p.add(l);
    p.add(Box.createHorizontalStrut(10));
    p.add(v);
    return p;
}

// --- About Us Page ---
private void showAboutUsPage() {
    frame.getContentPane().removeAll();
    JPanel mainPanel = new JPanel(new GridBagLayout());
    mainPanel.setBackground(new Color(252, 250, 246));
    GridBagConstraints gbc = new GridBagConstraints();
    gbc.gridx = 0;
    gbc.gridy = 0;
    gbc.weightx = 1.0;
    gbc.weighty = 1.0;
    gbc.anchor = GridBagConstraints.CENTER;
    gbc.fill = GridBagConstraints.BOTH;

    JPanel cardPanel = new JPanel();
    cardPanel.setLayout(new BoxLayout(cardPanel, BoxLayout.Y_AXIS));
    cardPanel.setBackground(Color.WHITE);
    cardPanel.setBorder(new CompoundBorder(
        new LineBorder(new Color(202, 97, 61), 2, true),
        new EmptyBorder(40, 40, 40, 40)
    ));
    cardPanel.setMaximumSize(new Dimension(700, Integer.MAX_VALUE));
    cardPanel.setAlignmentX(Component.CENTER_ALIGNMENT);
```

```
JLabel title = new JLabel("About StayZy");
title.setFont(new Font("Georgia", Font.BOLD, 32));
title.setForeground(new Color(202, 97, 61));
title.setAlignmentX(Component.CENTER_ALIGNMENT);
cardPanel.add(title);
cardPanel.add(Box.createVerticalStrut(20));

cardPanel.add(sectionHeader("Welcome to StayZy"));
cardPanel.add(sectionText("Your smart, comfortable, and hassle-free stay solution. Whether you're traveling for work, leisure, or a quick city stop, StayZy offers thoughtfully designed rooms and homestays with a focus on simplicity, comfort, and convenience."));
cardPanel.add(Box.createVerticalStrut(10));
cardPanel.add(sectionHeader("Our Mission"));
cardPanel.add(sectionText("To provide a reliable, modern stay experience at an affordable cost. We believe that traveling should be stress-free, and your accommodation should enhance your journey."));
cardPanel.add(Box.createVerticalStrut(10));
cardPanel.add(sectionHeader("What We Offer"));
cardPanel.add(sectionText("• Comfortable Rooms & Homestays\n• Free Wi-Fi, daily housekeeping, 24/7 support, and complimentary breakfast.\n• Strategic locations near major destinations.\n• Flexible & Transparent Booking — free cancellation, no hidden fees."));
cardPanel.add(Box.createVerticalStrut(10));
cardPanel.add(sectionHeader("Our Values"));
cardPanel.add(sectionText("Simplicity – We make travel frictionless.\nTransparency – Clear pricing and honest information.\nComfort – Clean, cozy, and reliable stays.\nTrust & Safety – Secure properties and verified hosts."));
cardPanel.add(Box.createVerticalStrut(10));
cardPanel.add(sectionHeader("Why Choose StayZy?"));
cardPanel.add(sectionText("• Cost-effective yet quality.\n• Flexible stays — short or long-term.\n• Local touch in our homestays.\n• Responsive customer support."));
cardPanel.add(Box.createVerticalStrut(10));
cardPanel.add(sectionHeader("Our Commitment"));
cardPanel.add(sectionText("We constantly improve our services based on guest feedback. From faster Wi-Fi and upgraded interiors to smoother booking experiences, StayZy evolves to make every stay better."));

JScrollPane scrollPane = new JScrollPane(cardPanel);
scrollPane.setBorder(null);
scrollPane.getVerticalScrollBar().setUnitIncrement(16);
scrollPane.setHorizontalScrollBarPolicy(ScrollPaneConstants.HORIZONTAL_SCROLLBAR_NEVER);
scrollPane.setVerticalScrollBarPolicy(ScrollPaneConstants.VERTICAL_SCROLLBAR_AS_NEEDED);
scrollPane.getViewport().setOpaque(false);
scrollPane.setOpaque(false);
mainPanel.add(scrollPane, gbc);
```

```
    JPanel bottomPanel = new JPanel();
    bottomPanel.setBackground(new Color(252, 250, 246));
    JButton backBtn = new JButton("Back");
    backBtn.setBackground(new Color(202, 97, 61));
    backBtn.setForeground(Color.WHITE);
    backBtn.setFocusPainted(false);
    backBtn.addActionListener(e -> showHomePage());
    bottomPanel.add(backBtn);
    gbc.gridy = 1;
    gbc.weighty = 0;
    mainPanel.add(bottomPanel, gbc);

    frame.add(mainPanel);
    frame.revalidate();
    frame.repaint();
}

// --- Helper methods for section headers and text ---
private JLabel sectionHeader(String text) {
    JLabel header = new JLabel(text);
    header.setFont(new Font("Georgia", Font.BOLD, 20));
    header.setForeground(new Color(202, 97, 61));
    header.setAlignmentX(Component.LEFT_ALIGNMENT);
    header.setBorder(new EmptyBorder(15, 0, 5, 0));
    return header;
}
private JLabel sectionText(String text) {
    JLabel label = new JLabel("<html>" + text.replace("\n", "<br>") + "</html>");
    label.setFont(new Font("Serif", Font.PLAIN, 15));
    label.setForeground(new Color(50, 50, 50));
    label.setAlignmentX(Component.LEFT_ALIGNMENT);
    label.setBorder(new EmptyBorder(0, 0, 10, 0));
    return label;
}

// --- Contact Page ---
private void showContactPage() {
    frame.getContentPane().removeAll();

    JPanel mainPanel = new JPanel(new BorderLayout());
```

```
mainPanel.setBackground(new Color(252, 250, 246));
mainPanel.setBorder(new EmptyBorder(30, 60, 30, 60));

// --- Card Panel for Contact Info ---
JPanel cardPanel = new JPanel();
cardPanel.setLayout(new BoxLayout(cardPanel, BoxLayout.Y_AXIS));
cardPanel.setBackground(Color.WHITE);
cardPanel.setBorder(new CompoundBorder(
   new LineBorder(new Color(202, 97, 61), 2, true),
   new EmptyBorder(30, 60, 30, 60) // More padding for better alignment
));

JLabel title = new JLabel("Contact Us");
title.setFont(new Font("Georgia", Font.BOLD, 32));
title.setForeground(new Color(202, 97, 61));
title.setAlignmentX(Component.CENTER_ALIGNMENT);
cardPanel.add(title);
cardPanel.add(Box.createVerticalStrut(10));

JLabel subtitle = new JLabel("We're here to help you — Reach out anytime!");
subtitle.setFont(new Font("Serif", Font.ITALIC, 18));
subtitle.setForeground(new Color(90, 90, 90));
subtitle.setAlignmentX(Component.CENTER_ALIGNMENT);
cardPanel.add(subtitle);
cardPanel.add(Box.createVerticalStrut(20));

cardPanel.add(sectionHeader("Email"));
cardPanel.add(sectionText("support@stayzy.com<br>reservations@stayzy.com"));

cardPanel.add(sectionHeader("Phone"));
cardPanel.add(sectionText("+91 98765 43210<br>+91 91234 56789<br>+91 99887 76655"));

cardPanel.add(sectionHeader("Address"));
cardPanel.add(sectionText("StayZy Pvt. Ltd.<br>45, Palm Avenue, Indiranagar,<br>Bengaluru, Karnataka, 560038, India"));

mainPanel.add(cardPanel, BorderLayout.CENTER);

JPanel bottomPanel = new JPanel();
bottomPanel.setBackground(new Color(252, 250, 246));
JButton backBtn = new JButton("Back");
```

```
    backBtn.setBackground(new Color(202, 97, 61));
    backBtn.setForeground(Color.WHITE);
    backBtn.setFont(new Font("Serif", Font.BOLD, 15));
    backBtn.setFocusPainted(false);
    backBtn.addActionListener(e -> showHomePage());
    bottomPanel.add(backBtn);
    mainPanel.add(bottomPanel, BorderLayout.SOUTH);

    frame.add(mainPanel);
    frame.revalidate();
    frame.repaint();
}

// --- Helper for navigation buttons ---
private JButton createNavButton(String text) {
    JButton button = new JButton(text);
    button.setBackground(new Color(252, 250, 246));
    button.setForeground(new Color(60, 60, 60));
    button.setFont(new Font("SansSerif", Font.PLAIN, 14));
    button.setBorderPainted(false);
    button.setFocusPainted(false);
    button.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
    return button;
}

private ArrayList<String> fetchAllLocations() {
    ArrayList<String> locations = new ArrayList<>();
    String query = "SELECT locationname FROM location ORDER BY locationname ASC";
    try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
         PreparedStatement stmt = conn.prepareStatement(query)) {
        ResultSet rs = stmt.executeQuery();
        while (rs.next()) {
            locations.add(rs.getString("locationname"));
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return locations;
}

// Debug utility: Print all hotel IDs from hotels and rooms tables
```

```
    private void debugPrintHotelAndRoomHotelIds() {
        System.out.println("[DEBUG] --- Hotel IDs in hotels table ---");
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             Statement stmt = conn.createStatement()) {
            ResultSet rs = stmt.executeQuery("SELECT hotelid, hotelname FROM hotels ORDER BY hotelid");
            while (rs.next()) {
                System.out.println("[DEBUG] hotels: hotelid=" + rs.getInt("hotelid") + ", hotelname=" +
rs.getString("hotelname"));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        System.out.println("[DEBUG] --- hotelid values in rooms table ---");
        try (Connection conn = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);
             Statement stmt = conn.createStatement()) {
            ResultSet rs = stmt.executeQuery("SELECT DISTINCT hotelid FROM rooms ORDER BY hotelid");
            while (rs.next()) {
                System.out.println("[DEBUG] rooms: hotelid=" + rs.getInt("hotelid"));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    public static void main(String[] args) {
        StayZyApp app = new StayZyApp();
        // Debug: print hotel IDs in both tables
        app.debugPrintHotelAndRoomHotelIds();
    }
}

class DateLabelFormatter extends AbstractFormatter {
    private String datePattern = "yyyy-MM-dd";
    private SimpleDateFormat dateFormatter = new SimpleDateFormat(datePattern);

    @Override
    public Object stringToValue(String text) throws ParseException {
        return dateFormatter.parseObject(text);
    }
```

```
@Override
    public String valueToString(Object value) throws ParseException {
        if (value != null) {
            Calendar cal = (Calendar) value;
            return dateFormatter.format(cal.getTime());
        }
        return "";
    }
}
```

Chapter 8

CONCLUSION

The Stayzy Hotel Room Booking System provides an efficient, user-friendly platform for managing hotel bookings and customer information. The system integrates key functionalities such as room availability management, user authentication, and detailed records of hotel locations and room types. By implementing database-driven operations with SQL and leveraging object-oriented programming in Java, Stayzy ensures accurate data handling and streamlined processes. Overall, the system enhances user experience, reduces manual errors, and serves as a reliable solution for hotel booking and administration.